

MUHAMMAD RAHMAN SHERAZI

Senior Infrastructure, AI & Platform Architect | Site Reliability Engineering (SRE) Lead

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PROFESSIONAL SUMMARY

Senior Infrastructure & Platform Architect with over 20 years of experience designing, optimizing, and scaling cloud-native ecosystems across AWS, GCP, and Azure. Expert at bridging deep engineering automation with high-level executive, security risk, and resilience visibility.

- **Platform & AI Engineering:** Pioneer in building custom Internal Developer Platforms (IDP) and AI/MLOps toolchains, utilizing proprietary DAG orchestration engines (coregraph), local RAG pipelines, and cross-platform IDE-to-browser ecosystems (SyncbridgeAI).
- **Automated Fleet Operations:** Expertise creating real-time incident orchestration engines (CO-Tooling-Bot) using Bash and Python to eliminate manual workflows, track service dependencies, and protect large-scale cloud fleets.
- **Enterprise Operational Intelligence:** Proven track record architecting cross-domain visibility solutions (KOIP) that map, track, and optimize compute, storage, network capacity, and core DNS infrastructure across multi-cloud and bare-metal environments.
- **SRE & Cross-Functional Leadership:** Deeply experienced in SRE disciplines, defining metrics (SLIs, SLOs, SLAs), managing error budgets, and leading 24/7 incident management to consistently maintain 99.9% platform uptime.

TECHNICAL SKILLS

- **Cloud Platforms:** AWS (EKS, IAM, EC2, S3), GCP, DigitalOcean, Azure (Azure DNS, Azure Networking, VNet, ExpressRoute).
- **DevOps & GitOps:** ArgoCD, Flux, Terraform, CloudFormation, Pulumi, Ansible, AWX, Helm, Bash Scripting.
- **Observability & SRE:** Prometheus, Grafana, Kibana, Splunk, PagerDuty, NetBox, Datadog, Metrics, Logging, Distributed Tracing, Synthetics, SLI/SLO Mapping, Service Health Dashboards.
- **Platform & AI/MLOps:** coregraph (Proprietary DAG Orchestration), KOIP (Operational Intelligence Platform), SyncbridgeAI (IDE-to-Browser Bridge), CO-Tooling-Bot (Incident Platform), OAR Renewals, JobMatch CLI/GUI, RAG Pipelines, Python, Ollama, ChromaDB.
- **Development Stacks:** Node.js (npm), TypeScript, JavaScript, Python, Go, PowerShell, FastAPI, Streamlit, Bash, SQL, VS Code Extension API, Browser Extension APIs.
- **Data & CI/CD:** MS SQL Server, PostgreSQL, SQLite, SQLAlchemy, Redis, MongoDB, GitHub Actions, GitLab CI, Jenkins, RunDeck.
- **Containerization & Orchestration:** Kubernetes (K8s), Docker, Rancher, OpenShift, CoreDNS, Route53, TCP/IP, Load Balancing, VPC/VNet Peering.
- **Security & Identity:** HashiCorp Vault, IAM, cert-manager, EKS, Container Security, Zero-Trust Architecture.
- **Strategy & Operations:** Capacity Management, Executive Reporting (CTO/CFO/Risk level), Disaster Recovery (DR), Business Continuity Planning (BCP), Operational Risk Tracking.

PROFESSIONAL EXPERIENCE

DigitalOcean | Cloud Operations Engineer II / SRE Lead

06/2024 – Present

- **Core Cloud Operations:** Manage and optimize high-availability bare-metal hypervisors, cloud infrastructure components, and distributed data systems across global data centers under strict enterprise SLAs.
- **Fleet Automation & Infrastructure as Code:** Designed and implemented Python- and Bash-based automation frameworks to orchestrate bare-metal lifecycle management, hypervisor administration, infrastructure validation, capacity discovery, and operational workflows across globally distributed cloud infrastructure.
- **Cloudy:** Modular CLI replacing fragmented dashboards and one-off scripts for DigitalOcean's CloudOps team — full droplet lifecycle management with peer-review-gated production changes and a second typed confirmation for irreversible actions, safety-verified REST API execution with crash-recoverable power-cycling, and a spare-capacity finder that flags unverifiable results rather than guessing.
- **CO-Tooling-Bot Orchestration:** Architected a real-time CloudOps incident response platform consolidating 8–12 parallel SSH terminals into a single CLI and web dashboard, with YAML-driven automated escalation (SSH → serial console → IPMI → BMC → PDU) and real-time integration with Atlantis, StackStorm, Slack, and PagerDuty.
- **Internal Product Development (coregraph):** Engineered an internal, proprietary DAG-based engine to map complex infrastructure endpoints, track service dependencies, and flag cross-resource vulnerabilities.
- **Operational Intelligence & Resilience (KOIP):** Spearheaded the architecture of KOIP, an infrastructure intelligence platform evolved from a Kubernetes resource optimizer into a multi-cloud cost and capacity intelligence suite, delivering automated savings snapshots to DevOps, CloudOps, FinOps, Billing, and CTO/CFO stakeholders.

- **Executive Visibility Dashboards:** Built interactive workspace monitoring dashboards using Python and Backstage APIs to provide risk, compliance, and engineering leaders with accurate, decision-ready service health scorecards.
- **Incident Management & SRE:** Participate in 24/7 on-call rotations, driving rapid resolution of infrastructure degradation, conducting deep post-mortems, defining SLIs/SLOs, and systematically reducing operational risk through self-healing automation.

SkyNet Software Inc. | Senior Administrator Infrastructure & Systems

06/2023 – 04/2024

- **AWS Orchestration:** Architected production-grade EKS clusters, focusing on high availability, disaster recovery failovers, and seamless node group scaling.
- **Security & IAM:** Applied least-privilege security by crafting granular AWS IAM roles and policies for service-to-service communication to enforce zero-trust compliance standards.
- **Cost Optimization:** Reduced cloud costs through EC2 rightsizing, Route53 optimization, storage lifecycle policies, and infrastructure modernization while improving availability and operational efficiency.
- **IaC Execution:** Accelerated cloud provisioning by 40% using reusable Terraform and Ansible modules integrated with automated validation, version control, and standardized deployment workflows.

Scotiabank | Tech Lead Data Integrations (Platform Engineering)

04/2022 – 01/2023

- **GCP & Azure Transformation:** Orchestrated zero-downtime migration of mission-critical Kubernetes applications and security-sensitive data services to Google Cloud Platform and hybrid cloud architectures while maintaining regulatory compliance.
- **Cloud Networking Engineering:** Designed secure hybrid networking including VPNs, routing, firewall policies, DNS integration, load balancing, and cloud connectivity between on-premises and public cloud environments.
- **Security Integration:** Implemented HashiCorp Vault and centralized secrets management to strengthen application security, credential lifecycle management, and automated secret rotation.
- **Hybrid Orchestration:** Managed hybrid CloudFormation and Terraform deployments through reusable Infrastructure-as-Code modules, automation pipelines, and governance controls.

OpenText | Lead Site Reliability Engineer

08/2021 – 04/2022

- **SRE Leadership:** Managed distributed microservices architectures using Docker and Kubernetes, tuning thresholds and defining error budgets to maintain a 99.9% platform uptime SLA.
- **Automation Stacks:** Built deployment validation pipelines using RunDeck, Helm, Kubernetes, and automated testing to improve release reliability and reduce production deployment risk.
- **Security Compliance:** Implemented Qualys vulnerability management and SentinelOne endpoint protection while improving monitoring, edge security, compliance reporting, and operational resilience.

Scotiabank | Infrastructure Consultant / Technical Specialist Advisory

04/2016 – 08/2021

- **Core Banking Ops:** Designed Terraform-based Infrastructure-as-Code solutions and orchestrated secure Kubernetes-based banking platforms across hybrid cloud environments while maintaining highly available network, storage, and compute infrastructure.
- **Leadership:** Managed cross-functional IT teams across Infrastructure, Network Operations, Cyber Security, and Compliance domains to eliminate reliability gaps and maintain strict uptime.
- **Compliance:** Ensured all hybrid-cloud architecture, alerting mechanisms, and operations dashboards adhered to strict financial regulatory, risk management, and data protection standards.

TECHNICAL INNOVATIONS & PRODUCT DEVELOPMENT

- **NearVar (Incident Runbook & Environment Panel):** Architected a persistent VS Code workspace utility consolidating local shell environments, AWS profiles, and markdown playbooks into a single sidebar dashboard. Engineered a fail-safe terminal stream layer that pastes scripts without automatic execution to eliminate human-error risks during critical on-call outages.
- **CO-Tooling-Bot (CloudOps Incident Response Platform):** Built with Python, FastAPI, Click, and asyncio, replacing 8–12 parallel SSH terminals and copy-paste workflows with a unified CLI and web dashboard — YAML-driven playbooks, automated escalation across SSH/serial console/IPMI/BMC/PDU, and real-time integration with Atlantis, StackStorm, Slack, and PagerDuty.
- **OAR Renewals (Service Catalog Operability Platform):** Automates Operational Availability Review compliance across 1,200+ microservices at DigitalOcean, replacing a manual, error-prone process: ingests the live service catalog hourly, validates documentation links, checks required operability metadata (PagerDuty, Jira, ownership), and produces per-service compliance scoring with monthly health snapshots via a read-only Streamlit dashboard. Built with Python, PostgreSQL, SQLAlchemy, Streamlit, Docker, Kubernetes, Vault.
- **KOIP (Infrastructure Intelligence Suite):** Started as a Kubernetes resource optimizer; evolved into a comprehensive platform mapping, analyzing, and optimizing cost structures across multi-cloud and on-premise compute, storage, and networking — serving DevOps, CloudOps, FinOps, Billing, and CTO/CFO teams with automated savings snapshots, without compromising SRE-grade operational safety.
- **coregraph:** A high-performance graph computation engine built for the KOIP platform to process complex knowledge-graph traversals at scale, modeling DAG-based infrastructure topology and dependencies.

- **SyncbridgeAI:** A cross-platform IDE-to-browser toolchain built using Node.js (npm) and TypeScript. It bridges local developer workspaces with web-based LLM chat interfaces by syncing states and execution hooks between a local VS Code extension and a custom Chrome extension.
- **JobMatch:** Self-hostable AI-powered job discovery and resume tailoring pipeline. Fetches roles from the Adzuna API and direct job boards, scores them against your resume via local TF-IDF and domain-overlap scoring, and generates ATS-tailored resumes through the Claude API — packaged with a FastAPI dashboard, Docker deployment, user accounts, and run-quota management.

EDUCATION & CERTIFICATIONS

- **Masters in IT (MSIT)** | PIMSAT
- Certified Kubernetes Administrator (CKA)
- AWS DevOps Engineer Professional
- AWS Solutions Architect Associate
- GitHub Copilot Certification
- VMware VCP (DC-DV)
- MCSE / MCITP
- MCP Azure
- CompTIA Network+
- CompTIA A+